

MANCHESTER AP, United Kingdom

WMO: 033340

Lat: 53.354N

Lon: 2.275W

Elev: 257

StdP: 14.56

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.0	28.2	20.0	15.1	30.2	22.7	17.3	31.9	30.8	45.8	27.6	46.2	3.3	90	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	78.5	64.5	74.8	62.9	71.5	61.4	65.8	74.4	64.3	72.1	62.8	69.5	8.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	86.3	69.1	61.1	81.3	67.1	59.8	77.5	65.6	30.7	74.7	29.5	72.3	28.4	69.6	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.4	21.5	19.1	DB	19.7	83.5	4.1	3.6	16.8	86.1	14.4	88.2	12.1	90.3	9.2	92.9	(4)
(5)			WB	19.1	69.4	4.1	2.3	16.2	71.0	13.8	72.4	11.5	73.6	8.6	75.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	40.2	40.9	43.6	47.8	53.2	58.0	61.7	61.2	56.9	51.5	45.0	40.8		
	DBStd	9.33	5.51	5.51	5.13	5.22	5.32	4.63	4.62	4.34	4.22	5.07	5.70	6.53		
	HDD50	1396	303	258	206	102	25	1	0	0	2	43	166	290		
	HDD65	5488	768	676	663	516	366	216	127	137	246	420	601	752		
	CDD50	1437	1	2	8	36	125	241	362	346	209	88	15	4		
	CDD65	53	0	0	0	0	1	7	24	18	3	0	0	0		
	CDH74	370	0	0	0	0	12	63	186	99	9	1	0	0		
	CDH80	57	0	0	0	0	0	7	30	20	0	0	0	0		
(14) Wind	WSAvg	8.9	10.2	10.0	9.9	9.0	9.1	8.2	7.8	7.8	8.0	9.0	9.0	9.2		
(15) Precipitation	PrecAvg	32.8	2.8	2.2	2.2	2.2	2.4	2.6	2.6	3.3	3.0	3.4	3.3	3.3		
	PrecMax	42.2	5.4	6.3	5.5	4.3	5.7	7.2	6.0	7.3	7.1	7.6	7.1	7.9		
	PrecMin	22.7	0.3	0.2	0.4	0.2	0.3	0.6	0.4	0.3	0.4	0.9	1.6	0.6		
	PrecStd	4.5	1.2	1.4	1.1	1.0	1.1	1.3	1.3	1.5	1.5	1.6	1.3	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	57.0	61.1	69.9	75.4	80.7	83.9	82.9	75.0	66.6	59.3	56.8		
		MCWB	51.6	50.9	51.8	56.9	61.3	65.3	66.1	65.5	63.3	60.3	56.0	53.3		
	2%	DB	52.1	53.4	57.1	63.5	71.4	75.0	78.9	76.0	70.1	62.9	57.2	54.0		
		MCWB	49.1	49.2	50.3	53.5	59.4	62.8	64.4	64.3	61.8	58.2	54.3	51.3		
	5%	DB	50.2	51.2	54.0	59.6	66.7	71.1	75.2	71.8	66.6	61.0	55.2	52.0		
		MCWB	47.5	47.6	48.8	51.5	56.9	60.6	63.4	62.6	59.4	57.0	52.2	49.5		
	10%	DB	48.4	49.3	51.9	57.0	62.8	67.7	71.5	69.0	64.4	59.1	53.2	50.0		
		MCWB	46.2	46.2	47.2	50.1	54.8	58.7	61.7	61.2	58.0	55.4	50.5	47.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	52.4	53.9	57.4	63.5	67.4	68.4	67.6	65.6	61.7	57.0	53.9		
		MCDB	54.7	55.2	58.7	65.0	73.3	76.5	78.0	76.6	72.0	65.7	59.1	56.1		
	2%	WB	50.0	50.0	51.5	54.8	60.1	63.8	65.9	65.5	63.1	59.5	54.8	52.2		
		MCDB	51.8	52.2	55.5	62.1	68.8	72.2	75.4	73.2	68.3	62.5	57.0	54.4		
	5%	WB	48.0	48.1	49.8	52.6	58.0	61.4	64.2	63.9	60.9	57.7	52.7	50.2		
		MCDB	49.6	50.7	53.4	58.8	65.6	68.9	72.7	70.5	65.0	60.5	54.8	52.0		
	10%	WB	46.2	46.3	48.0	50.6	55.8	59.5	62.5	62.4	59.3	55.8	50.9	47.9		
		MCDB	48.1	48.7	51.6	55.9	61.5	65.9	70.1	67.5	63.3	58.5	52.9	49.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	10.1	12.2	14.0	14.6	14.5	14.6	13.7	13.0	11.2	9.6	9.0		
		MCDBR	10.3	11.2	15.2	18.2	20.7	21.2	21.5	18.0	16.2	12.3	9.7	10.2		
		MCWBR	9.5	8.7	10.2	10.7	11.4	11.2	9.9	8.8	9.5	8.6	8.4	9.3		
	5% WB	MCDBR	9.9	10.6	13.7	16.8	18.5	18.3	18.3	15.5	14.1	11.3	9.6	10.0		
		MCWBR	9.7	9.0	10.1	10.6	10.9	10.9	9.7	8.6	9.5	9.0	8.9	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.339	0.369	0.395	0.398	0.397	0.397	0.382	0.367	0.349	0.324	0.313		
	taud		2.321	2.371	2.294	2.239	2.273	2.303	2.307	2.364	2.410	2.427	2.385	2.297		
	Ebn at Noon		202	241	256	264	270	271	268	266	255	234	202	181		
	Edh at Noon		19	25	33	39	40	39	38	34	29	23	18	16		
(44) All-Sky Solar Radiation	RadAvg		206	421	782	1209	1531	1602	1524	1252	904	505	251	158		
	RadStd		20	46	72	119	107	149	153	82	71	43	27	16		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (73 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page